Technische Universität Braunschweig



# Network Information Theory
# Channel Coding with State


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# Foundational Concepts:

## 1. Entropy (*H(X)*)

- Definition: A measure of uncertainty or randomness. randomness.

$$H(X) = -\sum_{x \in X} P_X(x) \log_2 P_X(x)$$

Entropy, *H(X)*, is a concept central to source coding. It quantifies the **inherent uncertainty** of a source. The set of '**typical**' sequences is a small subset of all possible sequences, and its size is approximately ***2^nH(X).***

## 2. Mutual Information

- Definition: The amount of of information shared between two random variables.

$$I(X;Y) = H(X) - H(X \mid Y)$$

Mutual Information, *I(X;Y)*. This is arguably the most crucial concept for concept for **channel coding**. It measures the amount of information information that the **output Y** provides about the **input X**. As you you can see from the formula, it's the the difference between the initial uncertainty of X and the remaining remaining uncertainty after observing observing Y. **Our goal is always to to maximize this value.**

## 3. Channel Capacity (C)

- Definition: The maximum maximum reliable transmission rate.

$$C = \max_{P_X} I(X;Y)$$

Channel Capacity, C, is the pinnacle of Shannon's work. It's work. It's the **maximum value value** of **mutual information information** over all possible possible input distributions. It distributions. It represents the the ultimate **physical limit** on on **reliable** communication for for a given channel.

## 4. Rate (R)

shows the efficiency of our coding.

It has 2 main goals:

1-In **Source coding**: Maximize the compression (**Minimize** the bit/symbol) rate.

2-In **Channel coding**: **Maximize** the message per symbol transition rate.

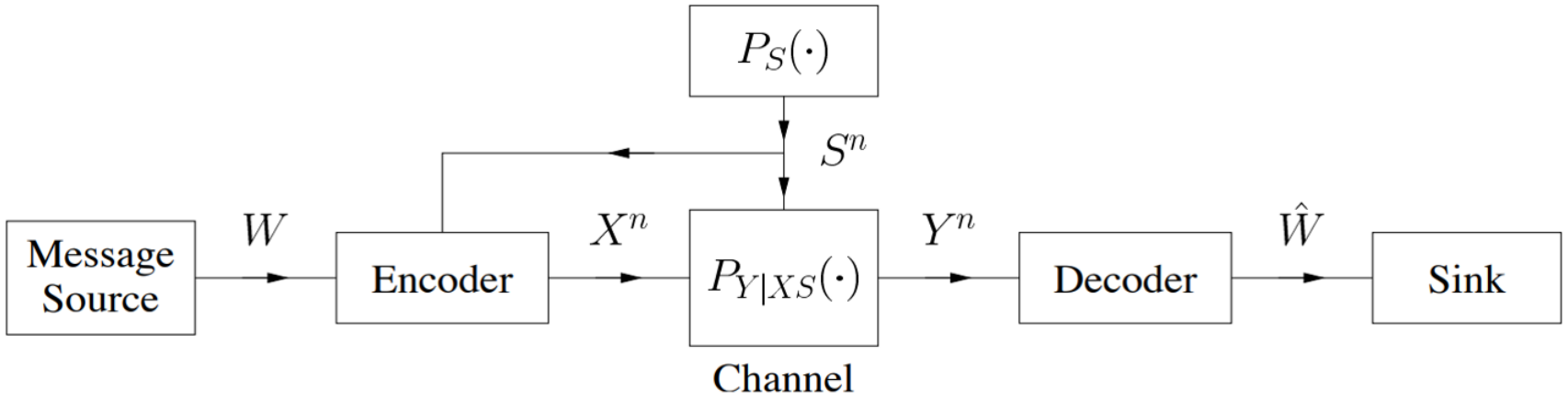
# Coding for Channels with State: The Gelfand-Pinsker Problem

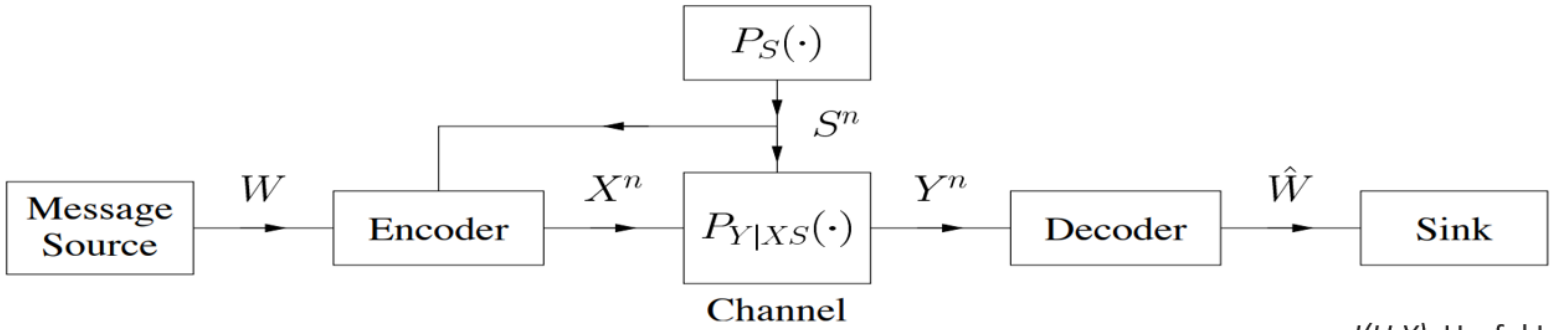
- **Central Question:** What happens if the **transmitter knows the channel's interference**, but the **receiver does not**?

- **Model Description:** A **Discrete Memoryless Channel** (DMC) influenced by a state sequence *Sn.*
- **Crucial Aspect:** The encoder has **non-causal access** to the entire state sequence *Sn* before encoding the message *W* into the channel input *Xn*. (past-now-future).
- **Input/Output Relationship:** The channel output *Yn* is a noisy function of the channel input *Xn* and the state *Sn*, typically represented by a conditional probability ***P(Y/X,S)*.**
- **Goal:** Maximize the achievable information rate *R* (bits per channel use) such that the probability of decoding error can be made arbitrarily small.

# The Gelfand-Pinsker Problem Model

- **Model Diagram:**
- **Transmitter:** Receives message *W* and state sequence *Sn*.
- **Encoder:** Produces channel input sequence *Xn* based on *W* and the known *Sn*.
- **Channel *P(Y/X,S),S(y/x,s)*** creates output *Yn*.
- **Receiver:** Receives channel output *Yn*.
- **Decoder:** Produces the decoded message *W^*.
- **Critical point:** Labeled '**State** *Sn*' points to the **Encoder** but **not to the Decoder**, illustrating that the knowledge of the state is available only at the transmitter.






$$C = \max_{p(u|s),x(u,s)} (I(U;Y) - I(U;S))$$

- *I(U;Y)*: Useful Information (information about message conveyed to conveyed to receiver).
- *I(U;S)*: The Cost (information shared between auxiliary variable and state, and state, which limits capacity).
- This implies a Markov chain U→S→X→Y

## S5.2 An Achievable Region

- Consider an auxiliary random variable $U$ with alphabet $\mathcal{U}$.
- Suppose $U$ is the output of a channel $P_{U|S}(\cdot)$.
- Define a function $f(\cdot)$ that maps symbols in $\mathcal{U} \times \mathcal{S}$ to $\mathcal{X}$, i.e., the sequence $x^n$ will have $x_i = f(u_i, s_i)$ for all $i$.
- We write the corresponding sequence mapping as $x^n = f^n(u^n, s^n)$.

**U depend on State**

**State**
**X depends on both S,U**

**Channel is memoryless**

**Code Construction:** Generate $2^{n(R+R')}$ codewords $u^n(w,v)$, $w = 1, 2, \ldots, 2^{nR}$, $v = 1, 2, \ldots, 2^{nR'}$, by choosing the $n \cdot 2^{n(R+R')}$ symbols $u_i(w,v)$ in the code book independently at random according to $P_U(\cdot)$.

**Binning**

# Binning in Gelfand-Pinsker Coding

- **What is it?**
- A technique for sending a message over a channel **with known interference (state).** It's like having a special **codebook that helps** the encoder deal with the interference **before the signal is even sent.**

- **How it Works (The Core Idea)**
- **Large Codebook**: The encoder creates a very **large collection** of codewords, **much larger than the number of messages.**

- **The "Smart" Encoder**: When the encoder wants to send a message, it looks at the interference and picks the *best* codeword from that message's bin. **This "best" codeword is one that is statistically compatible with the interference.**

- **The "Simple" Decoder**: Then the decoder just looks at the codewords which are statistically compatible with the received message and decodes the message.

**Analysis of the Probability of Error**

1. $\mathcal{E}_1 = \{s^n \notin T^n_{\epsilon_1}(P_S)\}$. The probability of this event approaches zero with $n$.
2. $\mathcal{E}_2 = \{s^n \in T^n_{\epsilon_1}(P_S)$ and $\nexists v : (u^n(w,v), s^n) \in T^n_{\epsilon_2}(P_{US})\}$.
   That is, the probability of this event is small if $\epsilon_2$ is small, $n$ is large and

$$R' > I(U;S). \tag{40}$$

3. $\mathcal{E}_3 = \{(u^n(w,v), s^n) \in T^n_{\epsilon_2}(P_{US})$ and $(u^n(w,v), y^n) \notin T^n_{\epsilon_3}(P_{UY})\}$. The probability of this event approaches zero with $n$.
4. $\mathcal{E}_3 = \{y^n \in T^n_{\epsilon_3}(P_Y)$ and the decoder finds a $(\tilde{w}, \tilde{v})$ with $\tilde{w} \neq w$ and $(u^n(\tilde{w}, \tilde{v}), y^n) \in T^n_{\epsilon}(P_{UY})\}$. That is, the probability of this event is small if $\epsilon$ is small, $n$ is large and

$$R + R' < I(U;Y).$$

**Solution:**

**1. N (goes up)[Large Numbers Law]**

**2. R' must be enough big to cover this situation.**

**3. N(goes up)[natural fault of channel]**

**4. R' must be minimized [R+R' must be limited]**

**WHY?**

•**State-Encoding Error (E1):** The encoder can't find a good way to encode the message because the channel's noise pattern is very unusual and unexpected.

•**Message-Decoding Error (E2):** The message gets scrambled by the channel's noise so badly that the decoder can't recognize the original message.

•**Message-Guessing Error (E3):** The decoder gets confused and picks the wrong message because its codeword looks too similar to the one that was actually sent.

•**State-Guessing Error (E4):** The decoder misinterprets the channel's noise pattern and, as a result, decodes the message incorrectly.

**OPTIMALITY Proof:**

**Converse**

We need:

**Theorem 8**

$$R(P_{SUXY}) = I(U;Y) - I(U;S)$$

*is convex in* $P_{X|SU}$.

The rate function is **convex**, has a **unique maximum**.
This guarantees that a single, optimal value for the channel capacity exists.

**Theorem 9 ((Data Processing Inequalities))** *If* $X - Y - Z$ *forms a Markov chain, then we have*

$$I(X;Z) \leq I(X;Y) \quad \textit{and} \quad I(X;Z) \leq I(Y;Z). \tag{41}$$

This principle states that processing data can't increase information.
The main benefit is providing an **upper bound**.

**Theorem 10 ((Fano's Inequality))** *Suppose both* $X$ *and* $\hat{X}$ *take on values in the alphabet* $\mathcal{X}$*, and let* $P_e = \Pr\{\{\}\hat{X} \neq X\}$*. We have*

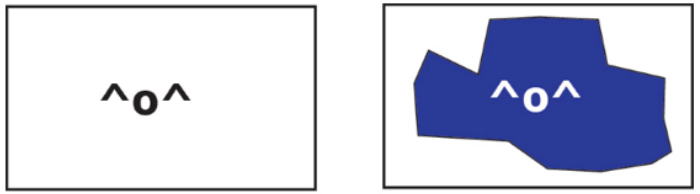
$$H_2(P_e) + P_e \log_2(|\mathcal{X}| - \infty) \geq \mathcal{H}(\mathcal{X}|\hat{\mathcal{X}}). \tag{42}$$

Fano's Inequality says that if the chance of a decoding **error is low**, then the remaining **uncertainty** about the original message **must be low**.

And we get:

$$R \leq \max_{P_{U|S}(\cdot),\, P_{X|SU}(\cdot)} I(U;Y) - I(U;S).$$

# Application of Gelfand Pinsker Problem: Dirty paper coding




Clean Channel  Dirty Channel

**Core Concept**: DPC is a technique for transmitting data reliably over a channel where the encoder knows a non-causal interference signal (the "dirty paper") in advance. It effectively **cancels out the known interference** at the transmitter, so the receiver only needs to decode the signal, not the combined signal and interference.

**Key Insight**: The encoder uses its knowledge of the interference to design a codeword that is "pre-subtracted" from the interference. This clever pre-coding ensures the combined signal-plus-interference at the receiver is simple to decode, as if the channel were clean.

**Main Result**: DPC achieves the same capacity as a clean channel with the same noise power, as if the interference didn't exist at all. The capacity is given by **C=maxP(U)I(U;Y)**, where U is an auxiliary variable chosen by the encoder.

# Conclusion: The Impact of Gelfand-Pinsker

•**Overcoming Interference:** The core achievement of Gelfand-Pinsker is showing how to **overcome known interference**. The encoder, by knowing the channel's "dirty paper" state, can use a smart technique called **binning** to pre-process the message.

•**Achieving Clean Channel Capacity:** This clever pre-processing allows the communication system to achieve the **same capacity as a clean, interference-free channel**. It proves that the presence of known interference doesn't have to reduce the communication rate.

•**Real-World Relevance:** This theory is not just an academic exercise. Its principles are the foundation for **Dirty Paper Coding (DPC)**, which is used in multi-user communication systems to manage and cancel out interference between different users. It highlights the immense value of side information at the transmitter.

Thanks for your time!